

Chapter 1

Bivariate Stable Functions and Directional Depth

1.1 Bivariate Stable Functions and Directional Depth

In this chapter we study bivariate independent stable random variables and explore the directional depth function in a new light. Recall that we consider a symmetric stable random variable

$$X \sim S(\alpha, 0, 1, 0)$$

with characteristic function

$$\varphi_X(t) = \exp(-|t|^\alpha).$$

Let X and Y be independent copies. Then, for any fixed constants a, b , the linear combination

$$Z = aX + bY$$

has characteristic function

$$\varphi_Z(t) = \exp\left[-\left(|a|^\alpha + |b|^\alpha\right)|t|^\alpha\right],$$

implying that

$$Z \sim S\left(\alpha, 0, c', 0\right), \quad \text{with} \quad c' = \left(|a|^\alpha + |b|^\alpha\right)^{1/\alpha}.$$

This scaling property lets us standardize the problem so that

$$P\left(aX + bY \geq au + bv\right) = P\left(\frac{aX + bY}{c'} \geq \frac{au + bv}{c'}\right).$$

For brevity we set

$$c = \frac{au + bv}{c'},$$

which represents the standardized directional threshold.

In our discussion the direction (u, v) is always taken as a unit vector (with respect to the chosen norm). This is analogous to the treatment in [1].

1.1.1 Analysis of the Directional Threshold Function and its Optimization

We study the directional threshold function defined by

$$c(a, b) = \frac{a u + b v}{(|a|^\alpha + |b|^\alpha)^{1/\alpha}},$$

where the direction (u, v) is assumed to be a unit vector and the weights (a, b) are chosen under the normalization, since $c(a, b)$ is homogeneous of degree 1.

$$|a|^\alpha + |b|^\alpha = 1.$$

A convenient parameterization is

$$a = t^{1/\alpha}, \quad b = (1 - t)^{1/\alpha}, \quad t \in [0, 1],$$

so that the function becomes

$$c(t) = u t^{1/\alpha} + v (1 - t)^{1/\alpha}.$$

Concavity versus Convexity. The nature of the mapping $x \mapsto x^{1/\alpha}$ is crucial:

- For $\alpha > 1$: Since $1/\alpha < 1$, the function $x^{1/\alpha}$ is *concave*. Thus, $c(t)$ is a concave function on $[0, 1]$, and its maximum is attained at a unique interior point.
- For $0 < \alpha \leq 1$: Here, $1/\alpha \geq 1$ so that $x^{1/\alpha}$ is *convex*; hence, $c(t)$ is convex on $[0, 1]$ and the maximum is attained at an endpoint.

Optimization for Different α Cases.

Case 1: $\alpha > 1$

Differentiating $c(t)$ and setting $c'(t) = 0$ leads quickly to

$$\left(\frac{1-t}{t}\right)^{\frac{\alpha-1}{\alpha}} = \frac{v}{u},$$

which yields the optimal value

$$t^* = \frac{1}{1 + \left(\frac{v}{u}\right)^{\frac{\alpha}{\alpha-1}}}.$$

Substituting t^* into $f(t)$ we obtain

$$c_{\max} = u \left[1 + \left(\frac{v}{u}\right)^{\frac{\alpha}{\alpha-1}} \right]^{\frac{\alpha-1}{\alpha}}.$$

Defining

$$\beta = \frac{\alpha}{\alpha - 1},$$

this simplifies to the elegant form

$$\boxed{c_{\max} = (u^\beta + v^\beta)^{1/\beta}}.$$

Case 2: $0 < \alpha \leq 1$

When $0 < \alpha \leq 1$ the function $c(t)$ is convex, and therefore its maximum is attained at one of the endpoints of the interval. Hence, in this case we have

$$c_{\max} = \max\{u, v\}.$$

1.2 Integral Representation for the Depth Function

Once the threshold $s_{\max}$ is determined, we evaluate the tail probability of a standard symmetric stable variable as follows. For

$$X \sim S(\alpha, 0, 1, 0),$$

the Gil–Pelaez inversion formula states that the CDF is

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin(tx)}{t} \exp(-t^\alpha) dt.$$

Thus, the tail probability becomes

$$P(X \geq x) = 1 - F(x) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\sin(tx)}{t} \exp(-t^\alpha) dt.$$

In our bivariate framework, after standardizing the linear combination with the scale factor $c' = (|a|^\alpha + |b|^\alpha)^{1/\alpha}$ and using the directional threshold $au + bv$, the probability of interest is

$$P(aX + bY \geq au + bv) = P(X \geq c),$$

where

$$c = \frac{au + bv}{c'}.$$

Plugging in the optimal threshold $s_{\max}$ (which now depends on the relative values of u and v and on α) we obtain the final formula for the depth:

$$\text{depth}(u, v) = P(aX + bY \geq au + bv) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\sin(t s_{\max})}{t} \exp(-t^\alpha) dt.$$

Here, note that:

- For $\alpha > 1$: $s_{\max} = (u^\beta + v^\beta)^{1/\beta}$ with $\beta = \frac{\alpha}{\alpha-1}$.
- For $0 < \alpha < 1$: $s_{\max} = \max\{u, v\}$.

1.3 Special Cases and Depth Contours and Visualization

For illustration, consider two scenarios:

- When $0 < \alpha < 1$ (say, $\alpha = 0.8$), the function $f(t) = u t^{1/\alpha} + v (1-t)^{1/\alpha}$ is convex so that the maximum is attained at one of the endpoints. Hence, the optimal threshold is

$$s_{\max} = \max\{u, v\}.$$

- When $\alpha > 1$ (say, $\alpha = 1.3$), the optimum is given by

$$s_{\max} = (u^\beta + v^\beta)^{1/\beta}, \quad \beta = \frac{1.3}{1.3 - 1} \approx 4.33.$$

Figure 1.1 shows the depth contours for $\alpha = 0.8$ and $\alpha = 1.3$. Notice that for $\alpha > 1$ the contours are given by the L^β -norm and tend to be rounder, while for $\alpha < 1$ the depth depends solely on the larger coordinate, leading to more “degenerate” contour shapes.

These examples also “prove” that our formulation is consistent with classical cases: for $\alpha = 1$ (the Cauchy case) one obtains $s_{\max} = \max\{u, v\}$ and a tail probability of

$$P(X \geq x) = \frac{1}{2} - \frac{1}{\pi} \arctan(x),$$

and for $\alpha = 2$ (the Gaussian case) with $\beta = 2$, the threshold becomes

$$s_{\max} = \sqrt{u^2 + v^2},$$

which is the familiar Euclidean norm. These special cases along with the contour plots vividly illustrate the rich geometric behavior of the depth function as α varies.

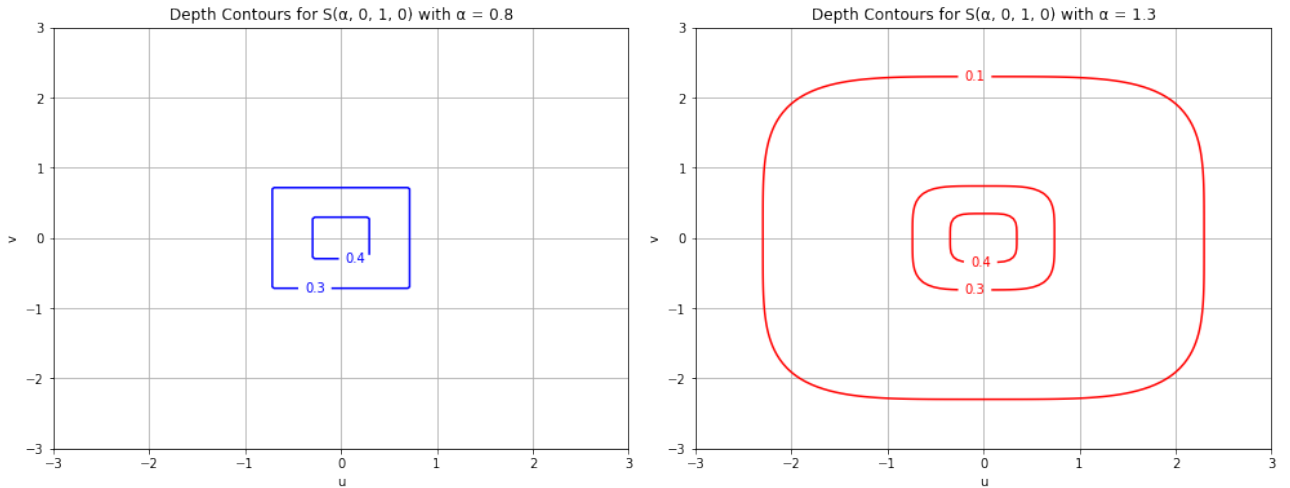


Figure 1.1: Depth contours for different values of α .

1.4 Optimal Directional Threshold in d Dimensions

Let $\mathbf{X} = (X_1, \dots, X_d)$ be a random vector with i.i.d. components $X_i \sim S_\alpha(0, 1, 0)$, and fix a unit direction $\mathbf{u} = (u_1, \dots, u_d)$, $\|\mathbf{u}\|_2 = 1$. For any weight vector $\mathbf{a} = (a_1, \dots, a_d) \neq \mathbf{0}$ define the *directional threshold*

$$c(\mathbf{a}) = \frac{\sum_{i=1}^d a_i u_i}{\left(\sum_{i=1}^d |a_i|^\alpha\right)^{1/\alpha}}.$$

Because both numerator and denominator are homogeneous of degree 1, we may normalise by imposing

$$\sum_{i=1}^d |a_i|^\alpha = 1.$$

Hence the optimisation problem becomes

$$\max_{\mathbf{a}} \sum_{i=1}^d a_i u_i \quad \text{s.t.} \quad \sum_{i=1}^d |a_i|^\alpha = 1.$$

Hölder-inequality solution ($\alpha > 1$)

Let $\beta = \frac{\alpha}{\alpha-1}$ so that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. By Hölder,

$$\sum_{i=1}^d a_i u_i \leq \left(\sum_{i=1}^d |a_i|^\alpha \right)^{1/\alpha} \left(\sum_{i=1}^d |u_i|^\beta \right)^{1/\beta} = \|\mathbf{u}\|_\beta,$$

and equality holds iff

$$|a_i|^\alpha = \frac{|u_i|^\beta}{\sum_{j=1}^d |u_j|^\beta}.$$

Taking the sign of u_i into account, the maximiser is

$$a_i^* = \frac{|u_i|^{\beta-1} \operatorname{sgn}(u_i)}{\left(\sum_{j=1}^d |u_j|^\beta \right)^{1/\alpha}}, \quad i = 1, \dots, d,$$

and the optimal value is the L^β -norm

$$c_{\max} = \|\mathbf{u}\|_\beta = \left(\sum_{i=1}^d |u_i|^\beta \right)^{1/\beta}.$$

Endpoint solution for $0 < \alpha \leq 1$

When $0 < \alpha \leq 1$ the map $x \mapsto |x|^\alpha$ is convex, so the constraint set is a convex body and the objective $\sum a_i u_i$ is linear. By convex-analysis the maximum occurs at an extreme point, i.e. where all but one $|a_i|$ are zero. Consequently

$$c_{\max} = \|\mathbf{u}\|_\infty = \max_{1 \leq i \leq d} |u_i|.$$

These two closed-form expressions for $c_{\max}$ feed directly into the one-dimensional Gil–Pelaez integral that defines the depth in any dimension d .

Bibliography

- [1] Rousseeuw, P. J., & Ruts, I. (1999). The depth function of a population distribution. *Metrika*, 235–236.